

# BHANU NAGPURE

Android Developer

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bhanu-dev82.github.io | Nagpur, India | Open to on-site / remote / hybrid

## SUMMARY

Android engineer with **2+ years** building and shipping production apps on Google Play. Deep expertise in **Kotlin, Jetpack Compose, and the Android SDK**, with a proven track record across **on-device AI** (LiteRT-LM, Gemma/Phi), custom camera pipelines (CameraX), MVVM + Hilt architecture, WorkManager and Play Store publishing. International research intern (Mitacs, Canada).

## TECHNICAL SKILLS

**Languages & UI:** Kotlin, Java, SQL; Jetpack Compose (Material 3), custom layouts, Material Design

**Architecture:** MVVM/MVI, Clean Architecture, Hilt (DI), Repository pattern, SOLID

**Concurrency & Persistence:** Coroutines & Flow, WorkManager, Room, Proto DataStore, EncryptedSharedPreferences, Retrofit, REST APIs

**On-Device AI:** LiteRT-LM (TFLite runtime), Gemma 2/3, Phi 3.5 Mini, MediaPipe FaceMesh, Google ML Kit

**Firebase & Monetisation:** Firebase (Auth, Firestore, Storage, Crashlytics), RevenueCat (IAP), Google AdMob mediation

**Tools:** Git, Gradle (Kotlin DSL), Android Studio, CI/CD, Google Cloud Platform, Linux

## PROFESSIONAL EXPERIENCE

### Devanshu Studios - Founder & Lead Android Engineer

Aug 2025 - Present

Independent · Remote

- Shipped **Chakuli - Offline AI Chatbot** (live on Google Play): a privacy-first Android assistant running **Gemma 2/3, Phi 3.5 Mini and SmolLM** entirely on-device via **LiteRT-LM** - zero cloud calls for AI inference; owned the full production lifecycle (CI/CD, Crashlytics, RevenueCat, AdMob mediation).
- Built the full **Jetpack Compose** (Material 3) UI, **Hilt DI**, **Proto DataStore**, **WorkManager** background model downloads, custom **CameraX** pipeline, Agent Skills system and encrypted key storage.
- Architected and shipped **PyMaster** - an Android/Flutter Python learning platform (50k+ LOC) with Gemini AI tutor, RevenueCat IAP and Firebase backend, live on Play Store.

### BETIC-GHRCE - Software Developer Intern

Oct 2022 - May 2024

Nagpur, India

- Built an **Android (Kotlin/Java)** app for TMJ-disorder screening integrating **MediaPipe FaceMesh** and **Google ML Kit**, improving diagnostic classification accuracy by **60%**.
- Boosted camera performance by **30%** via optimised ML Kit integration enabling real-time facial landmark detection - contributed to a filed patent (Pub. No. 202321022476).

### Wikimedia Foundation - Open Source Contributor

Feb 2025 - Present

Scribe-Android (Kotlin) · Remote

- Developed tablet-specific keyboard layouts in **Kotlin** for 7+ languages; resolved UI bugs and refactored the core **KeyHandler** module for maintainability and test coverage.

## KEY PROJECTS

### Chakuli - Offline AI Chatbot | Kotlin, Jetpack Compose, LiteRT-LM, Hilt, CameraX

2025 - Present

- On-device Android AI assistant running **Gemma 2/3, Phi 3.5 Mini, SmolLM** via **LiteRT-LM** - no internet required for inference, full data privacy; Agent Skills system, encrypted storage, foreground WorkManager model downloads, Protobuf DataStore persistence.

### OCanPredict - OSMF Screening App | Kotlin, Java, MediaPipe FaceMesh, Firebase

Dec 2022 - Jul 2023

- Android diagnostic app using **MediaPipe FaceMesh** for real-time OSMF screening, boosting efficiency by **40%**; won multiple national awards and contributed to a granted patent.

## EDUCATION

### G H Raisoni College of Engineering

Dec 2021 - Jun 2025

B.Tech in Computer Science & Engineering | CGPA: 8.7 / 10.0

Nagpur, India

Operating Systems, Computer Networks, DBMS, OOP, Data Structures & Algorithms

## PUBLICATIONS, PATENTS & AWARDS

- Journal:** "V-Safe-Anywhere: Empowering Women's Safety with Wearable AI and IoT." *IGI Global*, Mar 2024.
- Conference:** "Autonomous Floating Garbage Collection Device using Computer Vision and Robotics." *8th ICACS 2024, ACM, Hong Kong*, Oct 2024.
- Conference:** "Development of a Mobile Field Trip Application." *3rd Annual TRESL Lab International Symposium*, Athabasca, Canada, Jul 2024.
- Patent (Granted):** "Smart Chairside TMJ Examination & Muscle Activity Detector Tool," India (Pub. No. 202321022476).; **Patent (Pending):** "AI-Based Portable Device for Dental Disease Detection," DPIIT, India (App. No. 202421006731, Published Aug 2025).
- Awards:** 3rd Place · Avishkar 2024 | Runner-up · DIPEX 2023 & SHRISTI 2023 | BIREC E-YUVA Grant · Rs. 10 lakh seed funding | Winner · MEDHA 2022 & MEDIC-2022